DAY 08 • STUDENT HANDOUT

# Manual Checking and Verification

*Checking detections and actions yourself, patching security solution updates, and scanning infected systems*

| | |
|---|---|
| **Qualification** | Cyber Threat Monitoring Level I |
| **Unit of Competency** | CS-ICT251101 Monitor and Report Cyber Threats |
| **Coverage** | Element 3 Perform manual checking and verification: Performance Criteria 3.1–3.4; Range of Variables: Action; Required Knowledge 3.1 Log and detection management, 3.2 Security solution scan and operations procedure, 3.3 Knowledge in organization SOP; Required Skill 3.6 Verification skills |
| **Duration** | 8 hours (1 training day) |

## Learning Objectives

At the end of this session, you should be able to:

1. Check a detection yourself with its file hash, the console report and threat intelligence (PC 3.1).
2. Check the action of the security solution in its logs and on the host (PC 3.2).
3. Check security solution updates against the vendor bulletin and patch signatures as required (PC 3.3).
4. Use the security solution to scan an infected system, following the update-then-full-scan procedure (PC 3.4).
5. Record verified results and recognize failed actions that must be escalated.

## 1. Trust, but Verify

Security consoles summarise. A good analyst checks the facts behind the summary: is the file really malicious, did the action really work, are the signatures really current, and is the infected computer really clean? Element 3 of the Competency Standards turns these checks into four performance criteria. In the lab you follow SOC-SOP-04 Manual Checking and Verification (~/day8/soc_verification_sop.txt).

## 2. Check the Detection (PC 3.1)

A hash is a fingerprint of a file. SHA-256 turns any file into a 64-character code; change one byte and the code is completely different. Hashes let you check a detection without opening the file.

| Step (SOC-SOP-04) | How | Result |
|---|---|---|
| **V1.1 Compute the hash** | sha256sum <file> (Linux) | Get-FileHash <file> (Windows) |
| **V1.2 Compare with the console** | Different = the file changed after the detection | Verdict: hash mismatch |
| **V1.3 Look it up** | Threat intelligence list = verified malicious | Vendor known-good list = false positive; neither = unknown, escalate |

> **NOTE — Safe handling of samples**
> Never open, run or upload a suspicious file. Hashing reads the file without running it. In the lab the "samples" are harmless text files. In a real SOC, samples stay in a locked, isolated location, and only Tier 2 analyses them.

## 3. Check the Action (PC 3.2)

Microsoft Defender writes every action to its Operational event log. The event alone is not proof: check the host too.

| Event ID | Meaning |
|---|---|
| 1116 | Malware detected |
| 1117 | Action taken (Quarantine, Remove, Block) |
| 1118 | Action failed (non-critical), e.g. file in use |
| 1119 | Action failed (critical), e.g. access denied |
| 1000 / 1001 | Scan started / finished |
| 2000 | Security intelligence (signatures) updated |

- After Quarantine or Remove the file must be gone: Test-Path <file> returns False.
- Event 1117 but the file is still there = NOT VERIFIED. Events 1118 / 1119 = FAILED.
- Get-MpThreatDetection shows ActionSuccess (True/False) and ThreatStatusID (3 = quarantined, 102 = quarantine failed).
- Not verified or failed actions are escalated to Tier 2 (Day 9, PC 4.3).

## 4. Check and Patch the Updates (PC 3.3)

A security solution is only as good as its latest update. Compare three versions with the vendor bulletin:

| Version | What It Is | If Older Than the Bulletin |
|---|---|---|
| **Security intelligence (signatures)** | Descriptions of known threats; updated several times a day | Older: update now with Update-MpSignature |
| **Engine** | The scanning program that uses the signatures | Older: request the update from IT |
| **Platform** | The Defender product itself | Older: request the update from IT |

> **TIP — Verify the patch**
> After Update-MpSignature, check Get-MpComputerStatus (AntivirusSignatureVersion) and the Defender event 2000. An update you did not verify is an update you only hope happened.

## 5. Scan the Infected System (PC 3.4)

1. Update the signatures first, so the scan knows the newest threats.

2. Run a FULL scan: Start-MpScan -ScanType FullScan. A quick scan only checks the usual hiding places (memory, startup, Temp).
3. Verify what was found and what happened: Get-MpThreat, Get-MpThreatDetection, events 1001, 1116, 1117/1118, Test-Path.
4. Record the result. A failed action is escalated (Day 9).

Why the order matters: in Demo 3 a quick scan with yesterday's signatures found nothing, while the same computer, updated and fully scanned, had a trojan in AppData\Roaming. "No threats found" only means "nothing this scan knew about, in the places it looked."

## Key Terms

| Term | Meaning |
|---|---|
| Hash (SHA-256) | A fixed-length fingerprint of a file; any change gives a different hash |
| Threat intelligence | Lists of known-bad indicators (hashes, IPs, domains) shared by security teams and vendors |
| Known-good list | Hashes of genuine files published by their software vendor |
| Verification | Checking a result yourself with independent evidence |
| Signatures (security intelligence) | The definitions the antivirus uses to recognise threats |
| Quick scan / full scan | A scan of common hiding places / of every file on every drive |
| Quarantine | Moving a malicious file to a locked place where it cannot run |
| Failed action | The solution tried to act and could not: it must be escalated |

## References


- TESDA Competency Standards — Cyber Threat Monitoring Level I (ICT Sector), unit CS-ICT251101, Element 3.
- Microsoft Learn — Microsoft Defender Antivirus: Update-MpSignature, Start-MpScan, Get-MpThreatDetection and event IDs 1000–1119, 2000.
- Microsoft Learn — Get-FileHash; GNU coreutils — sha256sum.
- NIST FIPS 180-4, Secure Hash Standard (SHA-256).
- NIST SP 800-83 Rev. 1, Guide to Malware Incident Prevention and Handling for Desktops and Laptops.